

Android Application for an RC Charging and Discharging Circuit

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An RC circuit is an electric circuit consisting of a resistor and a capacitor. This article describes the simple RC charging and discharging circuit with an Android application as shown in Fig. 1. By using this Android application, you can calculate capacitor charging and discharging voltages and currents, and capacitor charging and discharging times with graphical outputs. The program is written in Java programming language. Achartengine-1.1.0

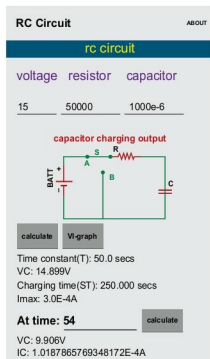


Fig. 1: RC Android application

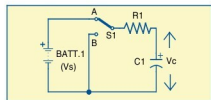


Fig. 2: Capacitor charging circuit

graph library is used to simulate the graph output.

RC charging circuit

The RC charging circuit is shown in Fig. 2. In this circuit, the resistor and capacitor are connected across the battery through mechanical switch S1. When switch S1 is closed, current from the battery passes through the resistor and, gradually, charges the capacitor until the capacitor voltage equals battery voltage ($V_c = V_s$).

At time $t=0$, the voltage gradually builds up across the capacitor as shown in Fig. 3 and the current

Calculation of various parameters

Time constant $\tau = R \times C$

Where τ = time constant, $R=R1$ =resistor and $C=C1$ =capacitor

Voltage across the charging capacitor $V_c = V_s(1 - e^{-t/RC})$

Where V_c =voltage across the charging capacitor, V_s =supply voltage, t =elapsed time

Voltage across the discharging capacitor $V_c = V_s(e^{-t/RC})$

RC =Time constant of the charging circuit

Current through the capacitor $I_c = \frac{V_s}{R} e^{-t/RC}$

Maximum current in the capacitor $I_{max} = \frac{V_s}{R}$

Where I_c =current through the capacitor, R =resistor, t =elapsed time and RC =time constant of the charging circuit

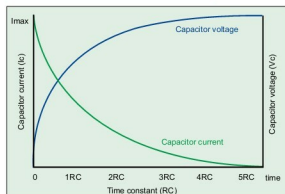


Fig. 3: A typical capacitor charging output graph

gradually decreases due to decrease in movement of electrons as shown in Fig. 3. The rate of charging is described in terms of time constant, denoted by $\tau = R \times C = R1 \times C1$. In the application, V_c is capacitor voltage, I_{max} is maximum current in the capacitor and I_c is capacitor current.

Formulae to calculate various parameters are given in the box.

For time $t = 1\tau$ (one time constant), $V_c = V_s(1 - e^{-1}) = 0.63V_s$

That is, the capacitor has been charged to 63 per cent of its final value. After 5τ (or five times constant), the capacitor will be fully charged.

For example,

If $R1 = R = 50k\text{-ohms}$, $V_s = 15V$, $C1 = C = 1000\mu F$, time constant $\tau = 50k \times 1000e^{-6} = 50.0$ seconds as shown in Fig. 1.

For time $t = 5\tau$, capacitor is fully charged. We need to calculate 5τ to know charging time t of the capacitor.

Using the formula given in the box, time $t = 5(50k \times 1000e^{-6}) = 250$ seconds, we get $V_c = 14.899V$, $I_{max} = 0.003A$ and $I_c = 2.0213\mu A$

At $t = 54$ seconds, $V_c = 9.906V$ and $I_c = 1.0187865e^{-4}$

RC charging output graph in the

Android application for the example is shown in Fig. 4.

In the output graph, the white line indicates the gradual decrease in current across the capacitor, and the yellow line indicates the gradual increase in voltage across the capacitor. This output waveform has one X axis and two Y axes. Left-side Y axis indicates the capacitor current